Unit: 6 Unit 4: Absolute Value
Topic: Definition of Absolute Value

Name: ________________
Date: ___________

Quiz (Carolina Reaper)

Q1. The absolute value of x represents the distance from x to:
(A) the origin on a coordinate plane
(B) the number 1 on a number line
(C) zero on a number line
(D) the number -1 on a number line
(E) the number pi on a number line

Q2. If a delivery truck is 5 miles away from a warehouse, the absolute value of its distance is:
(A) -5 miles
(B) 5 miles
(C) 10 miles
(D) 15 miles
(E) 20 miles

Q3. Evaluate the expression $|-2cdot3|$:
(A) -6
(B) 6
(C) -9
(D) 9
(E) 12

Q4. A shipment of packages is delayed by $|x-5|$ hours, where x is the scheduled arrival time. If $x=8$, the delay is:
(A) 3 hours
(B) 5 hours
(C) 7 hours
(D) 11 hours
(E) 13 hours

Q5. The equation $|x-2|=4$ has solutions $x=$
(A) $-2, 6$
(B) $-6, 2$
(C) $2, 6$
(D) $-2, -6$
(E) $6, -6$

Q6. A traffic signal has a cycle length of $|x+3|$ minutes, where x is the time of day. If $x=-5$, the cycle length is:
(A) 2 minutes
(B) 5 minutes
(C) 8 minutes
(D) 10 minutes
(E) 12 minutes

Q7. The inequality $|x|>3$ is equivalent to:
(A) $x>3$ or $x<-3$
(B) $x>3$ and $x<-3$
(C) $x<3$ or $x>-3$
(D) $x<3$ and $x>-3$
(E) $x=3$ or $x=-3$

Q8. The absolute value of $-3+4i$ is:
(A) 5
(B) $sqrt25$
(C) 1
(D) $sqrt3^2+4^2$
(E) $sqrt3^2-4^2$

Open Ended Questions (FRQ)

Q9. A company has two warehouses, A and B, which are $|x-10|$ miles apart. The company has a delivery truck that travels from warehouse A to warehouse B. If the truck travels at an average speed of $|2x+5|$ miles per hour, and $x=7$, how many hours will it take the truck to travel from warehouse A to warehouse B?

Q10. A shipping company has a fleet of trucks that travel from a central hub to various destinations. The company uses the equation $|d-200|=50$ to determine the distance d from the hub to each destination. Solve for d and explain the meaning of the solutions in the context of the shipping company.

Chef's Tips

- Ensure students understand the definition of absolute value as distance from zero on a number line.
- Remind students to consider both positive and negative solutions when evaluating absolute value expressions.
- Encourage students to interpret absolute value in contextual problems, such as transportation and logistics.